

Chapter22: 6, 18, 34, 47, 54, 61

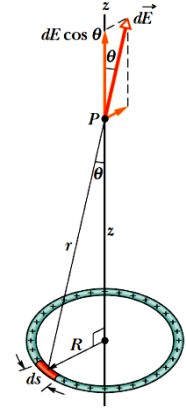
Basic score 4` + problem 18 2`+ problem 61 4`

6. As shown in figure, for a point at a distance z from the center of the uniform charged ring, the E field created by the ring at that point must be in z direction according to the symmetry of the charge. So we only need to calculate the z component of the field. Let the linear charge density of the charged ring to be $\lambda = Q/2\pi R$. The charge of the differential arc shown in figure is: $dq = \lambda dl = (Q/2\pi R)dl$, the z component of the its E field is:

$$dE_z = dE \cos \theta = \frac{dq}{4\pi\epsilon_0 r^2} \frac{z}{r} = \frac{z\lambda dl}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}}$$

So the net E field created by the ring can be get from integral:

$$E = \int_{\text{ring}} dE_z = \int_0^{2\pi R} \frac{\lambda dl}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}} = \frac{Qz}{4\pi\epsilon_0 (z^2 + R^2)^{3/2}}$$



(a) It is clear from symmetry that the field vanishes at the center. Or from above E field, when $z = 0$, we will get $E = 0$.

(b) The result ($E = 0$) for points infinitely far away can be reasoned directly from above expression (it goes as $1/z^2$ as $z \rightarrow \infty$)

(c) Differentiating the E field and setting equal to zero (to obtain the location where it is maximum) leads to

$$\begin{aligned} \frac{d}{dz} \left(\frac{Q}{4\pi\epsilon_0} z (z^2 + R^2)^{-3/2} \right) &= \frac{Q}{4\pi\epsilon_0} [(z^2 + R^2)^{-3/2} - 3z^2 (z^2 + R^2)^{-5/2}] = 0 \\ \Rightarrow 1 - \frac{3z^2}{z^2 + R^2} &= 0 \Rightarrow z = \frac{R}{\sqrt{2}} = 0.707R \end{aligned}$$

(d) Plugging this value back into E field expression with the values stated in the problem, we find $E_{\max} = 4.33 \times 10^7 \text{ N/C}$.

18. Examining the minimum value on the graph

$$U = -\vec{p} \cdot \vec{E} \rightarrow U_{\min} = -pE \cos 0^\circ = -100 \times 10^{-28} \text{ J}$$

or $pE \cos 0^\circ = 100 \times 10^{-28} \text{ J}$

1.5 points

If $E = 50 \text{ N/C}$, we find $p = 2.0 \times 10^{-28} \text{ C}\cdot\text{m}$. 0.5 point

34. For it to be possible for the net field to vanish at some $x > 0$, the two individual fields (caused by q_1 and q_2) must point in opposite directions for $x > 0$. Given their locations in the figure, we conclude they are therefore oppositely charged. Further, since the net field points more strongly leftward for the small positive x (where it is very close to q_2) then we conclude that $-q_2$ is the negative-valued charge. Thus, q_1 is a positive-valued charge. We write each charge as a multiple of some positive number q (not determined at this point). Since the problem states the absolute value of their ratio, and we have already inferred their signs, we have $q_1 = 4q$ and $-q_2 = -q$. The net electric field at $x > 0$ should be

$$E_{\text{net}} = E_1 + E_2 = \frac{4q}{4\pi\epsilon_0(L+x)^2} - \frac{q}{4\pi\epsilon_0 x^2}$$

for points along the positive x axis. Setting $E_{\text{net}} = 0$ at $x = 10.0$ cm (see graph) immediately leads to $L = x = 10.0$ cm.

(a) If we differentiate E_{net} with respect to x and set equal to zero (in order to find where it is maximum), we obtain (after some simplification) that location:

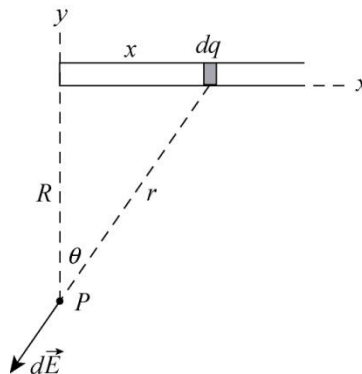
$$\frac{d}{dx} \left(\frac{4}{(L+x)^2} - \frac{1}{x^2} \right) = -8(L+x)^{-3} + 2x^{-3} = 0 \Rightarrow x = \frac{L}{4^{1/3} - 1} = 1.70L = 17.0 \text{ cm}$$

We note that the result for part (a) does not depend on the particular value of q .

(b) Now we are asked to set $q = 3e$, where $e = 1.60 \times 10^{-19}$ C, and evaluate E_{net} at the value of x (converted to meters) found in part (a). The result is 8.75×10^{-8} N/C.

47. Consider an infinitesimal section of the rod of length dx , a distance x from the left end, as shown in the following diagram. It contains charge $dq = \lambda dx$ and is a distance r from P . The magnitude of the field it produces at P is given by

$$dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{r^2}.$$



The x and the y components are

$$dE_x = -\frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{r^2} \sin \theta$$

and

$$dE_y = -\frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{r^2} \cos \theta,$$

respectively. We use θ as the variable of integration and substitute $r = R/\cos \theta$, $x = R \tan \theta$ and $dx = (R/\cos^2 \theta) d\theta$. The limits of integration are 0 and $\pi/2$ rad. Thus,

$$E_x = -\frac{\lambda}{4\pi\epsilon_0 R} \int_0^{\pi/2} \sin \theta d\theta = \frac{\lambda}{4\pi\epsilon_0 R} \cos \theta \Big|_0^{\pi/2} = -\frac{\lambda}{4\pi\epsilon_0 R}$$

and

$$E_y = -\frac{\lambda}{4\pi\epsilon_0 R} \int_0^{\pi/2} \cos \theta d\theta = -\frac{\lambda}{4\pi\epsilon_0 R} \sin \theta \Big|_0^{\pi/2} = -\frac{\lambda}{4\pi\epsilon_0 R}.$$

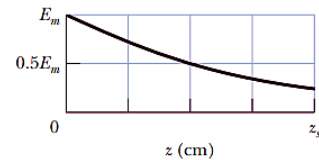
We notice that $E_x = E_y$ no matter what the value of R . Thus, $\vec{E}$ makes an angle of 45° with the rod for all values of R .

The net electric field is the vector sum of x and y component

$$E = \sqrt{E_x^2 + E_y^2} = \frac{\sqrt{2}\lambda}{4\pi\epsilon_0 R} = 1512 \text{ N/C}$$

54. From the lecture or textbook (page 572), we know the electric field created by the uniform charged disk is:

$$E = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{(z^2 + R^2)^{1/2}}\right) = E_m \left(1 - \frac{z}{(z^2 + R^2)^{1/2}}\right)$$



and note that this ratio is $\frac{1}{2}$ (according to the graph shown in the figure) when $z = 8.0$ cm.

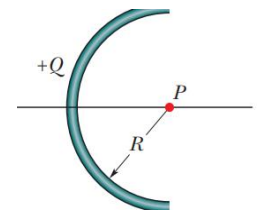
That is:

$$1 - \frac{8.0\text{cm}}{(8.0\text{cm}^2 + R^2)^{1/2}} = \frac{1}{2} \Rightarrow R = \sqrt{3}z = 13.9 \text{ cm}$$

61. Consider a differential element having arc length ds , the (infinitesimal) charge on an element ds of the rod contains charge $dq = \lambda ds$ (0.5 point). The element produces a differential electric field at the origin: $dE = \frac{1}{4\pi\epsilon_0} \frac{\lambda ds}{r^2}$. 0.5 point (这两条式子不要求单独

写出, 后面公式正确或者后面公式包含即给分)

From the symmetry analysis, we know the electric field sets up by the arc is along the symmetry axis.



$$E_{arc} = \int_{arc} dE = \int_{-90^\circ}^{90^\circ} \frac{\lambda R d\theta}{4\pi\epsilon_0 R^2} \cos\theta = \frac{\lambda}{4\pi\epsilon_0 R} (2 \sin 90^\circ) = \frac{\lambda}{2\pi\epsilon_0 R} = \frac{Q}{2\pi^2 \epsilon_0 R^2} = \frac{2kQ}{\pi R^2}$$

2 points

For point charge, the electric field at point P is

$$E = \frac{Q}{4\pi\epsilon_0 R^2}$$

0.5 point

So the ratio of the two E fields is

$$\frac{E_{particle}}{E_{arc}} = \frac{\frac{Q}{4\pi\epsilon_0 R^2}}{\frac{Q}{2\pi^2 \epsilon_0 R^2}} = \frac{\pi}{2} \approx 1.57.$$

0.5 point